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A candidate SiO jet tracing the outburst history of the FUor-type star V346 Nor

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Abstract

We report collimated, knotty SiO ($J = 5$–4) emission in archival ALMA Band 6 observations of the FU Orionis-type (FUor) young eruptive star V346 Nor. The emission is one-sided and redshifted, extends ∼780 au south-east of the source, and is resolved into four knots along an axis largely misaligned with the known ∼10 000 au CO outflow. The knotty, wedge-shaped structure rooted at the source and the velocity offsets that increase outward favour a newly launched jet. For jet speeds of 100–250 km s^{-1}, the kinematic ages of the knots place the ejection of the two outermost knots during the well-documented 1980–2010 outburst and that of the two knots closest to the source near its end or during the subsequent fading and recovery. If the jet interpretation is correct, V346 Nor is a rare case in which individual jet knots can be matched to a monitored accretion history. Five molecular transitions show a coherent velocity gradient perpendicular to the jet, consistent with rotation in a near-edge-on inner disk of ∼200 au diameter and an enclosed mass of ∼0.3 $M_\odot$. The inner-disk axis is largely misaligned with the axis of the large-scale pseudo-disk and CO outflow, possibly reflecting outburst-driven changes in the inner-disk geometry. Multi-epoch SiO imaging can measure the knot proper motions and thereby test the jet interpretation and remove the jet velocity and inclination ambiguities that remain in the present single-epoch data.

Keywords: stars: formation — stars: jets — stars: individual (V346 Nor) — ISM: jets and outflows — ISM: molecules

1. Introduction

FU Orionis-type objects (FUors) are young stars undergoing large-amplitude outbursts driven by sharp increases in disk-to-star accretion (Hartmann & Kenyon 1996; Audard et al. 2014; Fischer et al. 2023). Accretion in embedded protostars is universally accompanied by ejection in the form of winds, outflows, and collimated jets (Frank et al. 2014; Bally 2016), and chains of jet knots are widely read as fossil records of past accretion variations (e.g., Raga et al. 1990; Lee et al. 2015). FUors offer a rare experiment. Their accretion histories are documented by decades of photometric monitoring, so ejecta launched during an outburst can in principle be matched to the accretion event that produced them.

In practice this experiment has hardly been possible. CO molecular outflows are found in many FUors (e.g., Kóspál et al. 2017b; Takami et al. 2019; Cruz-Sáenz de Miera et al. 2023), and collimated jets have been reported in only some cases (e.g., Z CMa, Poetzel et al. 1989; L1551 IRS 5, Fridlund et al. 2005; SVS 13A, Hodapp & Chini 2014; V346 Nor, Kóspál et al. 2020). During an outburst the accretion ram pressure compresses the stellar magnetosphere and reconfigures the magnetic field near the launching region (Königl et al. 2011; Hartmann et al. 2016), but how this affects collimated ejection remains an open question. SiO is the standard tracer of shocked gas in Class 0/I protostellar jets (e.g., Codella et al. 2007; Lee et al. 2017a; Dutta et al. 2022), and among young eruptive stars SiO jet knots have been resolved in SVS 13A, whose outburst characteristics are intermediate between the FUor and EXor classes (Hodapp & Chini 2014; Lefèvre et al. 2017). Direct comparisons between resolved ejecta and a documented accretion history have nonetheless remained scarce.

V346 Normae (V346 Nor) is a southern FUor in the Sa 187 cloud at 700 pc (Reipurth 1985). Its outburst began in the early 1980s (Graham & Frogel 1985; Reipurth 1985) and peaked around 1992 (Ábráhám et al. 2004). Around 2010 the source faded rapidly, with the accretion rate dropping by two to three orders of magnitude (Kraus et al. 2016; Kóspál et al. 2017a), and it re-brightened within a few years (Kóspál et al. 2017a, 2020). ALMA 1.3 mm observations revealed a ∼10 000 au envelope with a CO outflow (Kóspál et al. 2017b) and a compact disk containing ∼0.18 $M_\odot$ of material, esti-

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mated from radiative-transfer modelling of the 1.3 mm continuum (Kóspál et al. 2021). VLT/X-SHOOTER spectroscopy taken in 2015 detected forbidden lines and H_2 emission suggesting a jet newly launched after the outburst (Kóspál et al. 2020).

Here we report collimated, knotty SiO emission in archival ALMA Band 6 data of V346 Nor, largely misaligned with the previously known large-scale outflow. We argue that a newly launched jet is its most likely origin. Section 2 describes the data. Section 3 presents the morphology and kinematics of the SiO emission, the knot properties, and the inner disk traced by complex organic molecules (COMs). Section 4 examines the nature of the SiO emission, relates the knot ejection epochs to the accretion history, and discusses the mass-loss rate, the inner disk, and its misalignment with the large-scale structure. Section 5 summarises the results.

2. Observations

V346 Nor was observed with the ALMA 12-m array in Band 6 on 2017 August 25, as part of the Cycle 4 project 2016.1.00209.S (PI M. Takami). For a full description of the observations we refer to Kóspál et al. (2021). Here we describe only the continuum, SiO, and molecular line data used in this paper. Two 1.8 GHz spectral windows centred at 216.9 and 232.2 GHz cover the 1.3 mm continuum, SiO ($J = 5$–4) at 217.105 GHz, and the five molecular transitions analysed in Section 3.3, namely CH_3OH at 216.946 and 217.299 GHz, CH_2DOH at 216.130 GHz, D_2CO at 231.410 GHz, and CH_3OCH_3 at 231.988 GHz.

We calibrated the data with the standard ALMA pipeline in CASA and imaged the continuum and the line cubes with *tclean* using Briggs weighting with robust $= -0.5$. This yields synthesised beams of $\sim 0''.14 \times 0''.09$, an rms noise of 0.067 mJy beam^{-1} for the continuum, and $\sim$1.5 mJy beam^{-1} ($\approx$3.0 K) per 0.7 km s^{-1} channel for the line cubes. To recover the faint SiO emission we additionally imaged the SiO cube with natural weighting at a velocity resolution of 1 km s^{-1}, obtaining a $0''.18 \times 0''.12$ beam and an rms noise of 1.4 mJy beam^{-1} (1.7 K) per channel. The SiO emission is detected at up to $\sim 8\sigma$ in this cube.

We also use the archival pipeline-imaged cube of ^{12}CO (2–1) at 230.538 GHz taken in the most extended configuration of the project, with a synthesised beam of $0''.14 \times 0''.09$. The cube binned to the 1 km s^{-1} channels of the natural-weighting SiO cube has an rms noise of 2.9 mJy beam^{-1}.

V346 Nor was also observed with the immersion grating infrared spectrograph IGRINS ($R \simeq 45\,000$; Park et al. 2014) on Gemini South in 2018 **[Jae-Joon Lee will update this part]**. The slit was oriented east–west through the central source, under seeing of $\sim 0''.5$.

3. Results

3.1. SiO Morphology, Kinematics, and Knot Properties

In Figure 1, the SiO ($J = 5$–4) emission shows a one-sided, redshifted structure extending south-east of the 1.3 mm continuum peak to a projected distance of $1''.1$ ($\sim$780 au at 700 pc), with no north-western counterpart. It is nearly perpendicular to the large-scale ($\sim$10 000 au) CO outflow, which runs from north-east to south-west (Kóspál et al. 2017b; Cruz-Sáenz de Miera et al. 2023), so the SiO emission does not share the axis of the previously known outflow. The peak-intensity map in Figure 2 resolves the emission into four knots, K1–K4, at projected distances of $0''.14$–$1''.11$ (98–777 au) from the continuum peak (Table 1).

The moment 1 map in Figure 1 shows that the emission becomes increasingly redshifted away from the source. We measure velocities relative to the systemic velocity $V_{\rm sys} = -5.0$ km s^{-1} obtained from CH_3OH at the continuum peak (Section 3.3). The peak velocity offsets of the knots increase outward, from $\sim$1.9 km s^{-1} at K1 to $\sim$8.7 km s^{-1} at K4 (Table 1). At larger scales Kóspál et al. (2017b) measured $V_{\rm sys} = -3.55$ km s^{-1} from C^{18}O with a finer velocity resolution. We keep -5.0 km s^{-1}, which centres the COM emission in Figure 5, and note that adopting -3.55 km s^{-1} would shift all velocity offsets in this paper by $+1.45$ km s^{-1}. This changes the velocity-related numbers slightly but not the outward increase of the offsets, the knot masses, or the qualitative interpretation of the SiO emission.

To examine how the knots are connected, we extracted a position–velocity (PV) diagram along the axis that joins them (PA $= 137°$, averaged over $0''.1$), passing through the continuum peak and K1, K2, and K4, while K3 lies slightly off-axis (Figure 2). The SiO emission is confined to within $\sim$10 km s^{-1} of $V_{\rm sys}$. Near the continuum peak it overlaps with the low-velocity CH_3OH ($E_{\rm up} = 55$ K) emission from the warm central gas (Section 3.3), showing that the SiO structure is rooted at the source. The knots do not form a single uniform chain, and their characters differ. K1 is compact and bright, and K2 is stretched in both position and velocity relative to the other knots, whereas the outer knots K3 and K4 are compact and kinematically isolated. Faint emission on either side of the axis between K1 and K2 forms a wedge that starts near K1 and opens away from the source, reaching a velocity extent of $\sim$10 km s^{-1} at K2 (green curve in Figure 2), giving an impression of emission entrained along the walls of a cavity.

The SiO spectra at the knot positions (Figure 3) are centrally peaked with extended wings and are not described by a single velocity component. We fitted each spectrum with a narrow plus a broad Gaussian (Table 1). The narrow components lie close to the local knot velocity with FWHM of 1–2 km s^{-1}, near the channel width, so the K3 and K4 widths are upper limits. The broad components are wider in K1 and K2 than in K3 and K4 by factors of $\sim$1.5–2.5, coinciding spatially with the wedge in the PV diagram. At our resolution the knots are at best marginally resolved, so the two components are a convenient description of the profiles rather than uniquely separated physical structures.

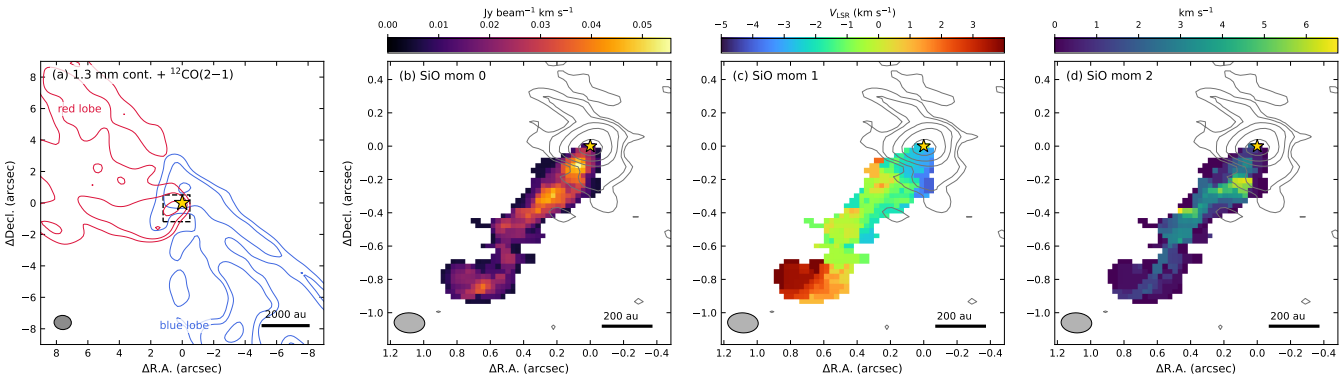


Figure 1. Overview of V346 Nor. In all panels the coordinates are offsets from the 1.3 mm continuum peak at $\alpha = 16^{\rm h}32^{\rm m}32^{\rm s}.197$, $\delta = -44°55'30''.77$ (J2000), marked by the star. *(a)* Large-scale view. Blue and red contours show the ^{12}CO (2–1) emission from Kóspál et al. (2017b), integrated over $1.5 < |V_{\rm LSR} - V_{\rm sys}| < 15$ km s^{-1} on each side, at 3, 5, and 10σ of each map. The dashed box marks the field shown in panels (b)–(d). *(b)–(d)* SiO ($J = 5$–4) moment 0 (integrated intensity), moment 1 (intensity-weighted velocity), and moment 2 (velocity dispersion) maps. Grey contours show the 1.3 mm continuum at 3, 5, 10, 20, and 50σ ($\sigma = 0.067$ mJy beam^{-1}). The synthesised beams (ellipses) and scale bars are shown in each panel. The SiO emission is redshifted, one-sided, and nearly perpendicular to the CO outflow.

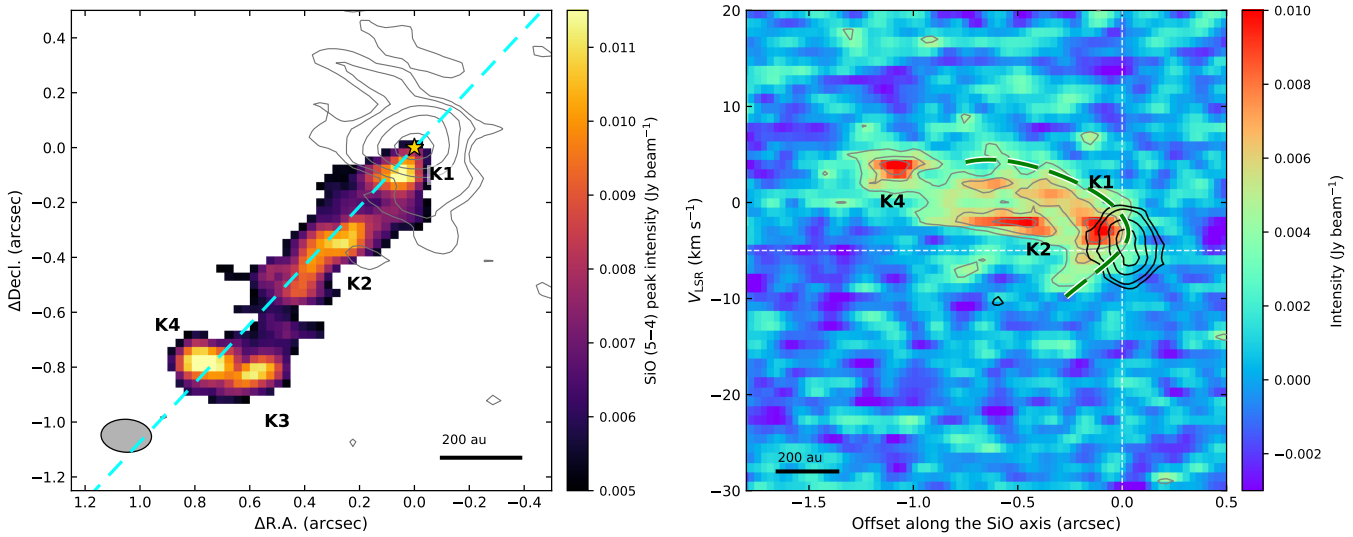


Figure 2. SiO peak-intensity map and position–velocity (PV) diagram. The left panel shows the SiO (5–4) peak-intensity (moment 8) map. The offsets, the star, the beam, the scale bar, and the grey continuum contours are as in Figure 1, and the knots K1–K4 are labelled. The cyan dashed line shows the PV cut (PA = 137°, averaged over 0$''$.1). The right panel shows the PV diagram along the cut. Grey contours show SiO at 2.5, 4, 6, and 9σ and black contours CH_3OH ($E_{\rm up} = 55$ K) at 3, 5, 10, and 15σ ($\sigma = 1.3$ and 1.6 mJy beam^{-1}). White dashed lines mark $V_{\rm sys} = -5$ km s^{-1} and the continuum peak. The green long-dashed curve outlines the wedge-like emission discussed in Section 3.1. K3 lies off-axis and is not sampled by the cut.

We estimated the SiO column density and mass of each knot from the moment maps, assuming optically thin LTE emission (Table 1). The peak brightness temperatures are 12.5–14.5 K. Since the knots are unresolved, these are beam-diluted lower limits on the excitation temperature. We adopt $T_{\rm ex} = 50$ K, within the range inferred for SiO in low-mass protostellar jets (e.g., Codella et al. 2007; Podio et al. 2021). A higher $T_{\rm ex}$ would raise the masses by a factor of a few.

The SiO column density follows from the integrated intensity as

$$N_{\rm SiO} = \frac{8\pi k_{\rm B} \nu^2}{hc^3 A_{ul}} \frac{Q(T_{\rm ex})}{g_u} \exp\left(\frac{E_u}{k_{\rm B} T_{\rm ex}}\right) \int T_{\rm b}\, dv, \qquad (1)$$

with $A_{ul} = 5.196 \times 10^{-4}$ s^{-1}, $g_u = 11$, $E_u/k_{\rm B} = 31.26$ K, and $Q \approx k_{\rm B} T_{\rm ex}/(hB)$, where $B = 21.712$ GHz is the rotational constant of SiO (Müller et al. 2005). This yields $N_{\rm SiO} = (2.9\text{–}8.2) \times 10^{13}$ cm^{-2}. Converting with a shock-enhanced abundance $X({\rm SiO}) = 10^{-7}$ (Ziurys et al. 1989; Schilke et al. 1997; Codella et al. 1999) gives knot masses of $(1.8\text{–}5.3) \times 10^{-6} M_\odot$, decreasing outward. These are order-of-magnitude estimates. For an unresolved knot the flux-based mass is nearly independent of the source size, and the dominant uncertainties are $T_{\rm ex}$ and $X({\rm SiO})$, the latter spanning 10^{-7}–10^{-6} in jets (e.g., Codella et al. 2007; Podio et al. 2021).

3.2. CO Emission around the SiO Chain

Figure 4 compares the CO and SiO emission. At $V_{\rm LSR} = -1$ to $+5$ km s^{-1} the CO forms a broad plateau south-east of

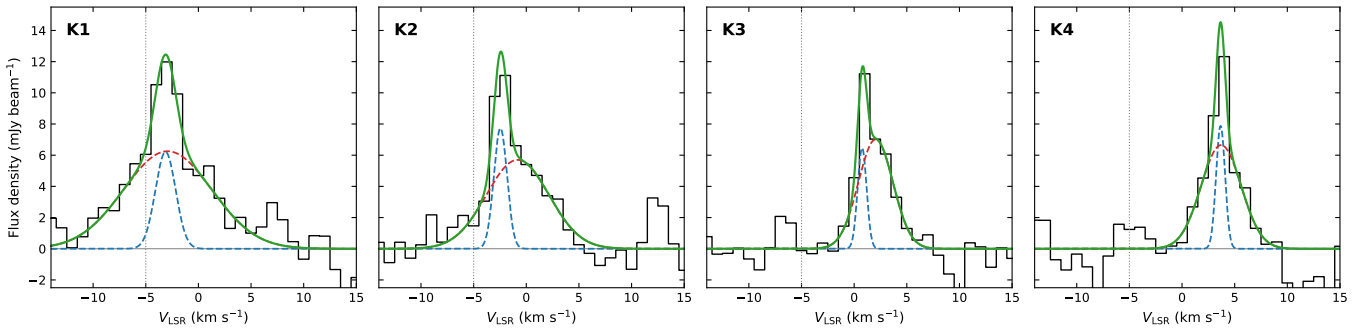


Figure 3. SiO ($J = 5$–4) spectra at the peak positions of knots K1–K4 (black histograms) with two-Gaussian fits, showing the narrow (blue dashed) and broad (red dashed) components and their sum (green solid). The dotted vertical line marks $V_{sys} = -5$ km s^{-1}. The narrow FWHM is constrained to 1–3 km s^{-1} (lower bound set by the 1 km s^{-1} channel width) and the broad FWHM to 3–10 km s^{-1}. Best-fit parameters are listed in Table 1.

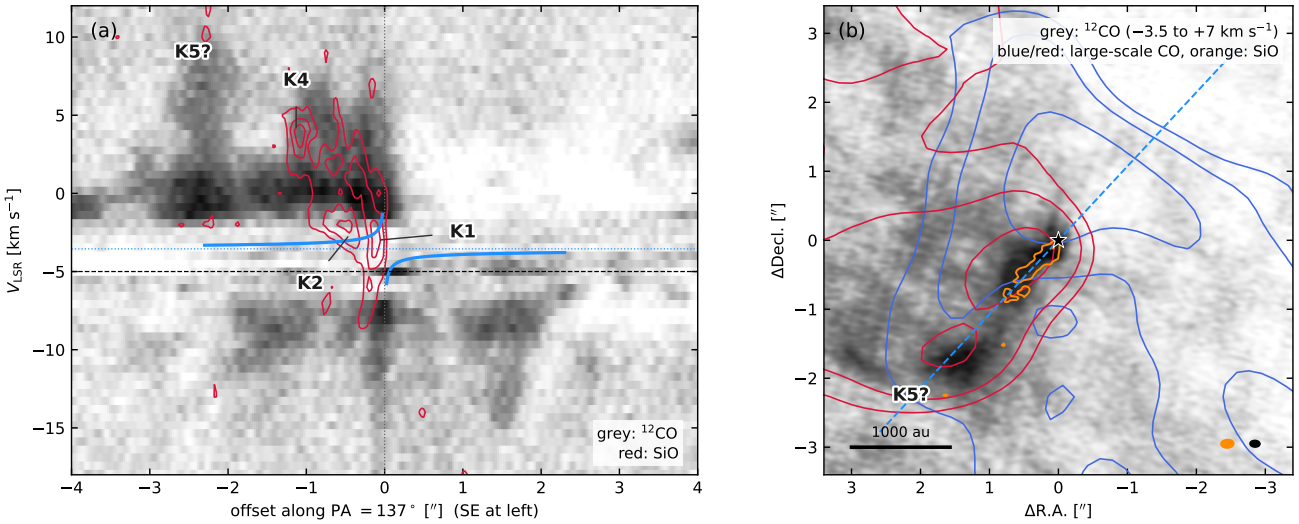


Figure 4. CO emission around the SiO chain. *(a)* ^{12}CO PV diagram (greyscale) along the SiO axis (PA = 137°, averaged over 0″.3), with the SiO PV emission from the 0″.1-wide cut of Figure 2 overlaid (red contours at the levels of Figure 2). The knots sampled by the cut are labelled, and K5? marks the compact high-velocity CO clump discussed in Section 4.1. The blue solid curves show the Keplerian rotation of the larger-scale structure of Kóspál et al. (2017b), for 0.1 $M_\odot$ about their systemic velocity of −3.55 km s^{-1} (blue dotted line). White horizontal bands near the systemic velocity are extended emission resolved out in the 12-m data. *(b)* ^{12}CO integrated over −3.5 to +7 km s^{-1} (greyscale), with the large-scale blue and red CO contours of Figure 1a, the SiO moment 0 (orange contours at 4, 8, and 16σ), and the sky position of K5? labelled. The blue dashed line marks the PV cut. The SiO chain runs along the south-western boundary of the redshifted CO plateau, and the redshifted CO continues along the cut beyond the chain.

the source, at the outer edge of the redshifted lobe of the large-scale outflow (Kóspál et al. 2017b). The SiO chain runs along the south-western boundary of this plateau, slightly outside it, and a lane of CO follows the chain at that boundary. Channel-by-channel maps at matched resolution are shown in Appendix A (Figure A.1).

The PV diagram along the SiO axis (Figure 4a) shows that the redshifted CO is not limited to the chain. Emission at −2 to +2 km s^{-1} continues to ∼3″.5 south-east of the source, and the velocity keeps rising beyond K4, up to a compact clump at ∼2″.3 and $V_{LSR} \approx +7$ km s^{-1} that we mark K5? in the figure. Blueshifted emission at −7 to −12 km s^{-1} appears on both sides of the source, consistent with the wide blueshifted lobe of Figure 1a crossing this region in projection.

The same panel also shows the Keplerian rotation of the larger-scale structure of Kóspál et al. (2017b). Beyond ∼100 au the Keplerian velocities stay within ∼1 km s^{-1} of their systemic velocity and decrease with distance, whereas the SiO emission lies at larger, mostly redshifted velocities that increase with distance. The SiO kinematics are unrelated to that rotating structure.

3.3. Inner Disk Traced by COM Emission

The CH_3OCH_3 ($E_{up} = 80$ K) emission forms a structure nearly perpendicular to the SiO axis, with a clear velocity gradient along its long axis (Figure 5), as also noted by Cruz-Sáenz de Miera et al. (2025). We adopt this direction, PA = 236°, as the major axis of the inner disk and extracted PV diagrams of five molecular transitions along it through the continuum peak.

All five transitions show a coherent gradient along PA = 236°, redshifted to the north-east and blueshifted to the south-

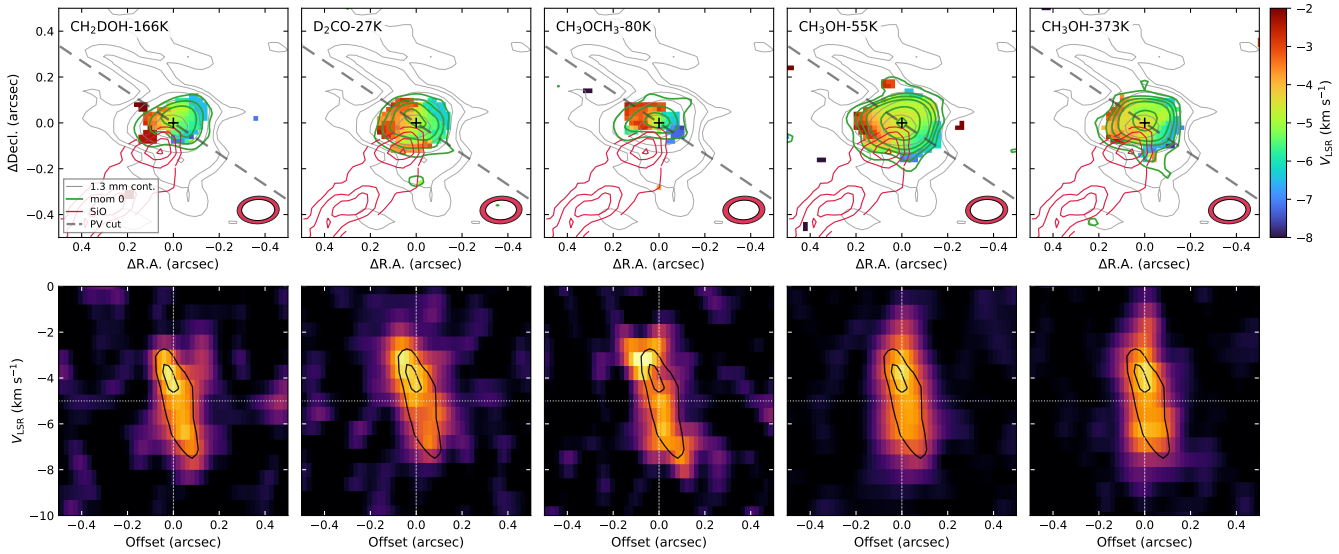


Figure 5. Inner-disk kinematics traced by five molecular transitions. The top row shows moment 1 maps of CH_2DOH ($E_{up} = 166$ K), D_2CO (27 K), CH_3OCH_3 (80 K), CH_3OH (55 K), and CH_3OH (373 K) over $V_{sys} \pm 3$ km s^{-1} ($V_{sys} = -5$ km s^{-1}). Grey contours show the 1.3 mm continuum at the levels of Figure 1, green contours the moment 0 of each line, and red contours the SiO moment 8 at 30, 50, 70, and 90% of its peak. The grey dashed line shows the PV cut (PA = 236°) through the continuum peak. Beams for the COM lines (white) and SiO (red) are at bottom right. The bottom row shows the PV diagrams along the cut in the same column order. Dotted lines mark V_{sys} and zero offset, and black contours show the CH_2DOH PV emission at 5, 7, 9, and 12σ for reference. The three optically thinner lines (D_2CO, CH_3OCH_3, CH_2DOH) share a narrow, quasi-linear ridge, whereas the two CH_3OH lines follow the same ridge plus additional optically thick emission concentrated toward the continuum peak.

Table 1. Properties of the SiO (5–4) knots in V346 Nor.

Knot	R.A., Decl. (J2000)	d ($''$)	d (AU)	T_b (K)	N^a_{SiO} (10^{13} cm^{-2})	M^a_{knot} ($10^{-6}\,M_\odot$)	t^b_{kin} (yr)	$\dot{M}^{a,b}_{jet}$ ($M_\odot$ yr^{-1})	v^c_0 (km s^{-1})	FWHM^c (km s^{-1})	A^c (mJy bm^{-1})	$\Delta V^{c,d}$ (km s^{-1})
1	16:32:32.20, −44:55:30.9	0.14	98	14.1	8.2	5.3	3.1	1.7×10^{-6}	-3.1 ± 0.3 -2.9 ± 0.6	2.3 ± 0.9 9.9 ± 2.6	6.2 ± 2.0 6.3 ± 1.8	$+1.9\pm0.3$ $+2.1\pm0.6$
2	16:32:32.22, −44:55:31.1	0.44	308	13.0	5.6	3.6	9.7	3.7×10^{-7}	-2.5 ± 0.2 -0.8 ± 0.7	1.5 ± 1.0 7.0 ± 1.3	7.7 ± 2.8 5.7 ± 1.5	$+2.5\pm0.2$ $+4.2\pm0.7$
3	16:32:32.25, −44:55:31.6	1.03	721	12.5	2.9	1.8	22.8	8.1×10^{-8}	$+0.8\pm1.5$ $+2.0\pm0.6$	≤ 1.0 3.9 ± 0.9	6.5 ± 17.7 7.0 ± 2.7	$+5.8\pm1.5$ $+7.0\pm0.6$
4	16:32:32.27, −44:55:31.5	1.11	777	14.5	3.5	2.3	24.6	9.3×10^{-8}	$+3.7\pm1.5$ $+3.7\pm0.4$	≤ 1.0 4.5 ± 1.5	7.9 ± 43.8 6.7 ± 3.7	$+8.7\pm1.5$ $+8.7\pm0.4$

For each knot, the upper and lower entries in the v_0, FWHM, A, and ΔV columns correspond to the narrow and broad Gaussian components, respectively.
[a]Order-of-magnitude estimates (Section 3.1).
[b]Kinematic ages for $V = 150$ km s^{-1} (Section 4.2).
[c]Best-fit two-Gaussian parameters.
[d]$\Delta V \equiv v_0 - V_{sys}$ with $V_{sys} = -5$ km s^{-1}.

west (Figure 5). The three optically thinner lines, D_2CO ($E_{up} = 27$ K), CH_3OCH_3 (80 K), and CH_2DOH (166 K), share a narrow, quasi-linear PV ridge reaching $|V - V_{sys}| \simeq 2.5$–3 km s^{-1} at $\pm0''.1$ (a radius of ∼70 au). The full extent of the COM emission is $\sim0''.3$, i.e., ∼200 au in diameter. The two CH_3OH transitions follow the same ridge plus additional optically thick emission concentrated toward the continuum peak. Coherent, quasi-linear velocity gradients of this kind, symmetric about the source and shared by several dense-gas tracers, are commonly interpreted as rotation in inner envelopes and disks (e.g., Goodman et al. 1993; Oya et al. 2016). A close analogue is HH 212, where COM emission traces a rotating disk atmosphere perpendicular to the SiO jet (Lee et al. 2017b).

We measured the ridge slope $\alpha = dV/dx$ of each PV diagram with a bisector fit. The method and per-line results are given in Appendix B. The three optically thin tracers give consistent slopes with a mean of $\alpha_{bis} = -20.8 \pm 3.1$ km s^{-1}/$''$ (Table B.1), whereas the two CH_3OH lines are biased by the optically thick component and are excluded. At $R = 0''.1$ (∼70 au), $V_{rot} = |\alpha_{bis}|R \simeq 2.1$ km s^{-1} and the enclosed mass is

$$M_{enc} = \frac{R V_{rot}^2}{G \sin^2 i_{disk}} \simeq 0.3\,M_\odot\,(\sin i_{disk})^{-2}, \qquad (2)$$

with a scatter of 0.25–0.5 $M_\odot$ for different transitions. For a near-edge-on disk ($i_{disk} \approx 90°$, consistent with the small line-of-sight velocities of the SiO knots), $M_{enc} \approx 0.3\,M_\odot$, consistent with the combination of the 0.1 $M_\odot$ stellar mass obtained from a Keplerian fit to the larger-scale structure (Kóspál et al. 2017b) and the ∼0.18 $M_\odot$ disk mass estimated from the dust

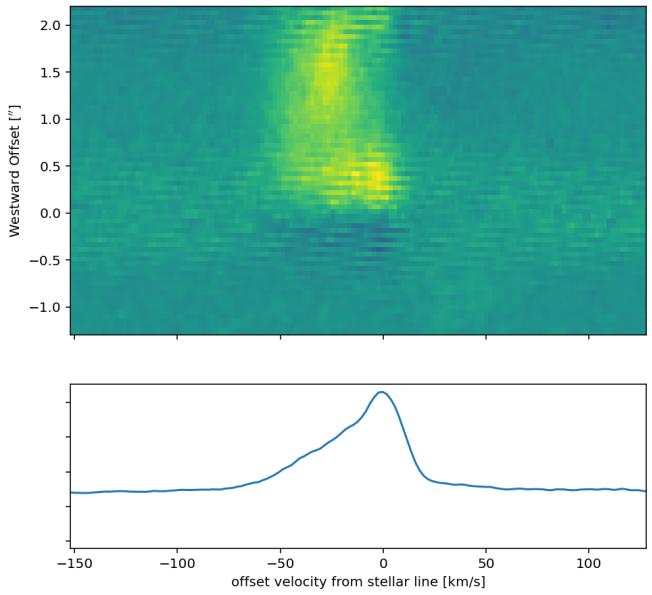


Figure 6. Gemini/IGRINS H_2 observations of V346 Nor along the east–west slit. *Top:* position–velocity distribution of the H_2 emission (abscissa: LSR velocity; ordinate: positional offset, east to west, across the continuum peak). Emission near the central star is subtracted in the top panel to emphasise the faint extended H_2 emission. *Bottom:* H_2 spectrum extracted within $0''.5$ of the central source. Blueshifted emission extends to $\sim 2''.5$ west of the source.

continuum (Kóspál et al. 2021), and with earlier indirect estimates of the stellar mass (Kraus et al. 2016; Kóspál et al. 2017a).

Because R is comparable to the beam ($\sim 0''.14 \times 0''.09$) and the gradient is sampled by only a few 1 km s^{-1} channels, $M_{\rm enc}$ is a beam-averaged value uncertain by at least a factor of a few. An inner Keplerian region, whose $V \propto R^{-1/2}$ spur would be beam-smeared into a quasi-linear ridge (e.g., Ohashi et al. 2014; Aso et al. 2015), cannot be distinguished from rigid-body or angular-momentum-conserving rotation at this resolution.

4. Discussion

4.1. The Nature of the SiO Emission

In protostellar systems SiO emission arises both in collimated jets and in shocked gas that the jet or outflow entrains (e.g., Codella et al. 2007; Gueth & Guilloteau 1999; Santiago-García et al. 2009; Podio et al. 2021). The SiO emission of V346 Nor could in principle be either. The chain runs along, and slightly outside, the boundary of the redshifted CO plateau (Section 3.2), where the outflow meets the surrounding gas, so SiO from material compressed at the outflow wall cannot be excluded.

The structure of the SiO emission, however, is the typical appearance of a jet. It is a straight chain of compact knots rooted at the source, with velocity offsets that increase outward, narrow line cores on broad wings, and a wedge that opens away from K1. These are the patterns produced by internal shocks in a jet and by the gas the jet itself entrains (e.g., Gueth & Guilloteau 1999; Santiago-García et al. 2009; Tafalla et al. 2017), whereas entrained cavity-wall gas usually forms spatially continuous arcs rather than kinematically isolated knots. The CO beyond the chain fits the same picture, with the outward velocity rise of the knots continuing into the CO up to the clump K5? (Section 3.2). With the current SiO and CO data alone we cannot distinguish whether the SiO traces the jet itself or entrained material.

Shock-related lines were absent toward the star in 1984–85, but a 1993 K-band spectrum showed H_2 emission, during the outburst (Graham & Frogel 1985; Reipurth et al. 1997; Kóspál et al. 2020), and the 2015 X-SHOOTER spectra showed blueshifted forbidden lines and H_2 emission from a newly launched jet, although its direction on the sky was not constrained (Kóspál et al. 2020). The IGRINS spectra provide a partial constraint (Figure 6). The H_2 emission is blueshifted, over a velocity range similar to the X-SHOOTER jet, and extends to $\sim 2''.5$ west of the source along the slit. The western side lies opposite the redshifted SiO knots with respect to the COM-traced disk axis, and the velocity sign is opposite to that of the SiO emission, as expected for the blueshifted counterlobe of a bipolar jet, although the single east–west slit does not follow the SiO axis and samples only a one-dimensional cut. The independent detection of an optical and near-infrared jet launched in this period leads us to expect a molecular jet as well.

A jet that is bright in SiO on one side and in optical or near-infrared lines on the other is not unusual. One-sided or strongly asymmetric jets are reported in other protostars and are explained by different extinction or ambient conditions on the two sides or by intrinsically asymmetric launching (e.g., Dutta 2026). Shock tracers can also differ between the two lobes of a single outflow, as in IRAS 15398−3359, where SiO and SO appear only at one spot of the outflow shell while JWST traces hot H_2 and [Fe II] elsewhere (Okoda et al. 2025). The redshifted SiO chain and the blueshifted optical and near-infrared emission of V346 Nor may be such a pair.

4.2. Ejection Epochs, Mass Loss, and the Accretion History

If each knot is a distinct ejection, its kinematic age is $t_{\rm kin} = d/V$ with $V \equiv v_{\rm jet} \cos i$, where i is the inclination of the jet from the plane of the sky. The small line-of-sight velocities of the knots indicate motion close to the plane of the sky, so we adopt $i \simeq 0°$. Figure 7 compares the inferred ejection epochs with the K_S light curve. For a fiducial value of $V = 150$ km s^{-1}, within the range typical of Class 0/I jets (e.g., Bally 2016), the ages are 3–25 yr (Table 1) and the ejection epoch of K4 coincides with the 1993 appearance of shock-tracing H_2 emission (Reipurth et al. 1997; Kóspál et al. 2020). K4 and K3 then fall in the outburst plateau ($\sim$1993 and 1995), K2 near its end ($\sim$2008), and K1 in the post-fade recovery ($\sim$2014). Across the 100–250 km s^{-1} range typical of Class 0/I SiO jets (Bally 2016; Podio et al. 2021) this grouping is robust. K3 and K4 were ejected during the 1980–2010 outburst, and

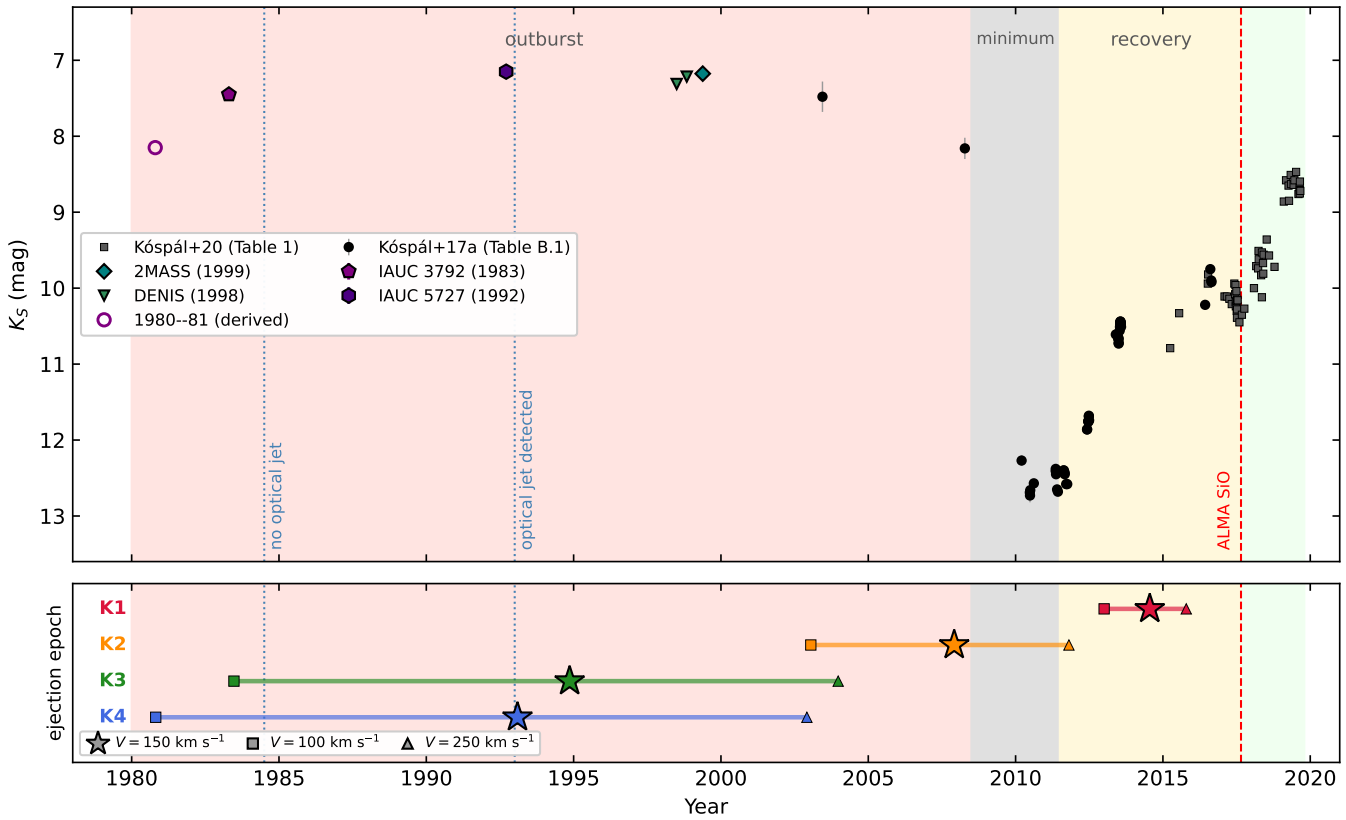


Figure 7. K_S-band light curve of V346 Nor and inferred knot ejection epochs. The top panel shows published photometry, taken from Kóspál et al. (2017a) (Table B.1, with NaCo, VISTA/VVV, IRIS, and SMARTS data), Kóspál et al. (2020) (Table 1, with SMARTS, IRIS, and VVVx data), 2MASS (Skrutskie et al. 2006), and DENIS (Epchtein et al. 1997), plus the 1983 measurement of Frogel & Graham (1983) and the 1992 measurement of Molinari et al. (1993). The open circle shows the 1980–81 level implied by Frogel & Graham (1983), who reported the source to be ∼0.7 mag brighter at 2.2 μm in 1983 than measured by Reipurth & Wamsteker (1983). In both panels the background shading marks the 1980–2010 outburst, the deep minimum, the post-fade recovery, and the renewed brightening (Kóspál et al. 2017a, 2020), dotted blue lines mark the 1984–85 spectroscopy that showed no jet signature and the 1993 first detection of shock-tracing H_2 emission (Reipurth et al. 1997; Kóspál et al. 2020), and the red dashed line marks the 2017.65 epoch of the ALMA SiO observations. The bottom panel shows the ejection epoch inferred for each knot. Stars mark $V \equiv v_{\rm jet} \cos i = 150$ km s^{-1}, and the horizontal bar of each knot spans the epochs for $V = 100$ (squares) to 250 km s^{-1} (triangles) (Section 4.2). For any velocity in this range K3 and K4 were ejected during the outburst, whereas the epochs of K1 and K2 shift with the adopted velocity.

K1 and K2 near its end or during the recovery. Multi-epoch proper motions would fix V directly. For the same velocities the kinematic age of the CO clump K5? is 30–50 yr, placing its ejection around the onset of the outburst.

The SiO profiles, a narrow core on broad wings, resemble those of internal working surfaces or bow shocks in Class 0/I jets, where the broad component traces shocked or laterally ejected gas and the narrow component less disturbed jet material (Codella et al. 2007; Tafalla et al. 2017; Jhan & Lee 2021). The narrow components have nearly constant amplitude across all four knots, while the broad components are strongest in K1/K2, consistent with the wedge-like feature (Section 3.1) tracing bow-shock wings and entrained ambient gas around the inner knots.

The apparent mass-ejection rates, $\dot{M}_{\rm jet} = M_{\rm knot}/t_{\rm kin}$, range from 8×10^{-8} to $1.7 \times 10^{-6} M_\odot$ yr^{-1} (Table 1). Their meaning depends on the knot-formation picture. For a discrete bullet (Norman & Silk 1979) it is not a launching rate and scales as $1/t_{\rm kin}$, whereas for an internal working surface (Raga et al. 1990; Lee et al. 2015) it approximates the time-averaged mass-loss rate. K1, with an age of only ∼3 yr and a morphology consistent with ongoing ejection, provides the most meaningful reference. Its $\dot{M}_{\rm jet} \approx 1.7 \times 10^{-6} M_\odot$ yr^{-1}, compared with the accretion rate at that time, $\dot{M}_{\rm acc} \sim 3 \times 10^{-6} M_\odot$ yr^{-1} (Kóspál et al. 2017a, 2020), gives a launching efficiency $\eta = \dot{M}_{\rm jet}/\dot{M}_{\rm acc} \sim 0.5$, or less if part of the broad component is entrained gas. Both values are within the standard Class 0/I range (e.g., Bally 2016; Lee 2020; Podio et al. 2021).

By contrast, the outburst-era knots K3 and K4 have SiO masses comparable to K1 despite an accretion rate that was then more than an order of magnitude higher ($\sim 10^{-4} M_\odot$ yr^{-1}, Kraus et al. 2016; Kóspál et al. 2017a). Several explanations are possible and not mutually exclusive. The launching efficiency may have been lower during the outburst, when the accretion ram pressure compressed the magnetosphere and altered the field geometry of the launch region (Königl et al. 2011; Hartmann et al. 2016), although whether such a change suppresses the ejection is untested. The outburst-era ejecta may instead have been largely atomic or ionised, as in the jet seen spectroscopically in 2015 (Kóspál et al. 2020), which

SiO does not trace. Finally, SiO production may have been lower during the outburst. SiO appears when shocks release Si from dust grains, and outburst heating can sublimate or reprocess the small grains of the inner disk, as observed for the crystallised silicates of EX Lup (Ábrahám et al. 2009; Kóspál et al. 2023) and EC 53 (Lee et al. 2026), so the ejecta launched during the outburst could carry less SiO.

4.3. The Inner Disk, its Snowline, and the Misalignment

The 1980–2010 outburst reached $L_{\rm bol} \approx 160\,L_\odot$ (Kóspál et al. 2017a,b). Since freeze-out timescales at these densities are 10^2–10^3 yr, ices sublimated during the outburst remain in the gas phase at the 2017 epoch, and the extent of the COM emission should be compared with the snowline at outburst peak. The water snowline follows $R_{\rm snow} \approx R_0\,(L_{\rm bol}/L_\odot)^{1/2}$ with $R_0 = 15.4$ au for an optically thin envelope, the value for a sublimation temperature of 100 K (Bisschop et al. 2007; van't Hoff et al. 2018). When a dense disk shields the midplane the scaling factor drops to roughly one third of this value, $R_0 \approx 5.1$ au (Kim et al. 2025). At $L_{\rm bol} = 160\,L_\odot$ these give ≈195 and 65 au. The observed ∼70 au radial extent of the COM emission is close to the disk-shielded value and well inside the envelope-only one, consistent with an inner disk that shields the midplane.

On larger scales, Kóspál et al. (2017b) identified a flattened rotating structure with a major axis at PA = 129°, Keplerian within ∼350 au, and the CO outflow runs roughly perpendicular to it (Kóspál et al. 2017b; Cruz-Sáenz de Miera et al. 2023). The inner disk discussed here (major axis PA = 236°) is oriented nearly perpendicular to that outer structure. Comparable inner/outer misalignments are reported in a growing number of embedded protostars and are attributed to precession in binaries, anisotropic infall, or episodic accretion reconfiguring the inner disk (e.g., Sakai et al. 2019; Matsushita et al. 2019; Okoda et al. 2021; Lin et al. 2024; Bate 2018; Kuffmeier et al. 2021). The large amplitude and short recurrence of the accretion history of V346 Nor favour an outburst-related origin, and independent evidence for outburst-driven geometric change exists. Kóspál et al. (2020) reported a deepening of the silicate absorption after the outburst and concluded that the geometry of the circumstellar material had changed, and outburst heating is known to reprocess the inner-disk dust (Ábrahám et al. 2009; Kóspál et al. 2023; Lee et al. 2026). Because the single-epoch data cannot constrain the 3D jet orientation, we regard the misalignment as suggestive rather than definitive. Knot proper motions and higher-resolution imaging of the inner-disk rotation are the direct tests.

5. Conclusions

Archival ALMA Band 6 observations of V346 Nor reveal a candidate newly launched SiO jet in this FUor.

1. The SiO (J = 5–4) emission forms a one-sided, redshifted, collimated structure extending ∼780 au southeast of the continuum peak, with four knots (K1–K4) and an axis (PA ≃ 137°) largely misaligned with the known ∼10 000 au CO outflow. Its knotty, wedge-shaped structure rooted at the source, its velocity offsets that increase outward, and the 1993 appearance of an optical jet at the epoch expected for the outermost knot favour a newly launched collimated jet over material entrained at the walls of the pre-existing outflow.

2. Under optically thin LTE assumptions ($T_{\rm ex} = 50$ K, $X({\rm SiO}) = 10^{-7}$), the knots have $N_{\rm SiO} = (2.9\text{–}8.2) \times 10^{13}$ cm^{-2} and order-of-magnitude masses of $(1.8\text{–}5.3) \times 10^{-6}\,M_\odot$. For $V = 150$ km s^{-1} the kinematic ages are 3–25 yr. Matched to the K_S light curve, K3 and K4 were ejected during the 1980–2010 outburst and K1 and K2 near its end or during the recovery, robustly across $V = 100\text{–}250$ km s^{-1}.

3. Five molecular transitions show a coherent velocity gradient along PA ≃ 236°, perpendicular to the jet, consistent with a near-edge-on rotating inner disk with $M_{\rm enc} \simeq 0.3\,M_\odot$. Its ∼70 au radial extent lies between the envelope-only (∼195 au) and disk-shielded (∼65 au) water snowlines expected at outburst peak, close to the latter.

4. The inner-disk axis is misaligned by several tens of degrees with the large-scale pseudo-disk/CO outflow, possibly reflecting outburst-driven changes in the inner-disk geometry.

The jet interpretation, the jet velocity, and the SiO-based mass-loss rates all carry uncertainties that single-epoch data cannot remove. Multi-epoch SiO imaging of the knot proper motions, together with higher-resolution observations of the inner disk, can settle them. For $V = 100\text{–}250$ km s^{-1} at 700 pc the knots move by 0″.03–0″.08 per year, so a second epoch taken within a decade would see them shift by two to five beam widths. If the jet picture is confirmed, V346 Nor will be a prime target for comparing the accretion and mass-loss rates across its outburst, fading, and re-brightening phases, and thus for studying how jets are launched during episodic accretion.

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Appendix A. CO Channel Maps

Figure A.1 compares the ^{12}CO and SiO emission channel by channel, using the CO data described in Section 2 at the matched 1 km s^{-1} channels and comparable beams. From -1 to $+5$ km s^{-1} the CO plateau grows south-east of the source and a lane of CO follows the SiO chain at its south-western boundary.

Appendix B. PV Ridge-Slope Measurement

We traced the PV ridge of each molecular transition in two complementary ways, using the intensity-weighted mean offset in each velocity channel and the intensity-weighted mean velocity in each offset column. Both tracings use pixels above $4\sigma_{\rm PV}$ within $|x| < 0''.3$ and $|V - V_{\rm sys}| < 4$ km s^{-1}, where $\sigma_{\rm PV}$ is measured from the source-free region. An ordinary least-squares fit to either tracing is biased steep or shallow depending on which variable is treated as independent, so we adopt the bisector of the two fits (Isobe et al. 1990). Figure B.1 shows the diagnostics and Table B.1 the slopes and enclosed masses. The three optically thin tracers give consistent bisector slopes (-20.0, -18.1, -24.2 km s^{-1}/$''$), whereas the two CH_3OH transitions are biased by the optically thick emission at $V_{\rm sys}$ and are excluded from the dynamical-mass estimate.

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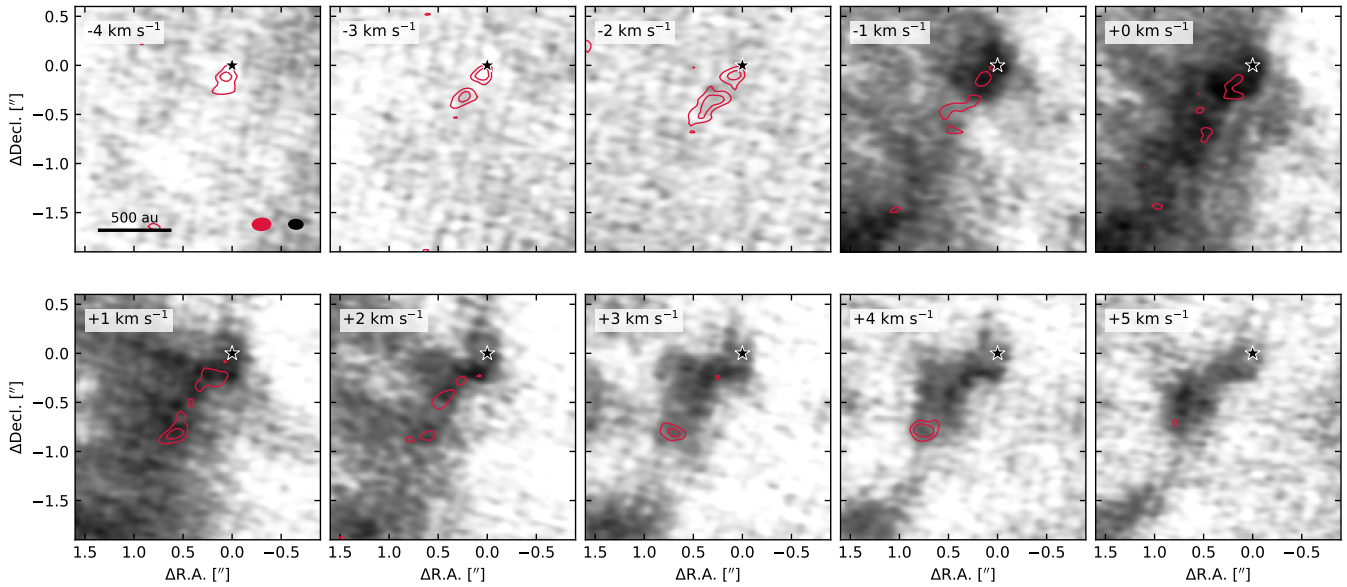


Figure A.1. ^{12}CO (2–1) channel maps (greyscale) compared with the SiO (5–4) emission (red contours at 4, 6, 10, and 15σ, σ = 1.5 mJy beam^{-1}) in matched 1 km s^{-1} channels. The central $V_{\rm LSR}$ of each channel is labelled, and the star marks the continuum peak. The filled CO (black) and SiO (red) beams and a scale bar are shown in the first panel. From -1 to $+5$ km s^{-1} the CO forms a plateau south-east of the source, and a lane of CO runs along the SiO chain at the south-western boundary of the plateau.

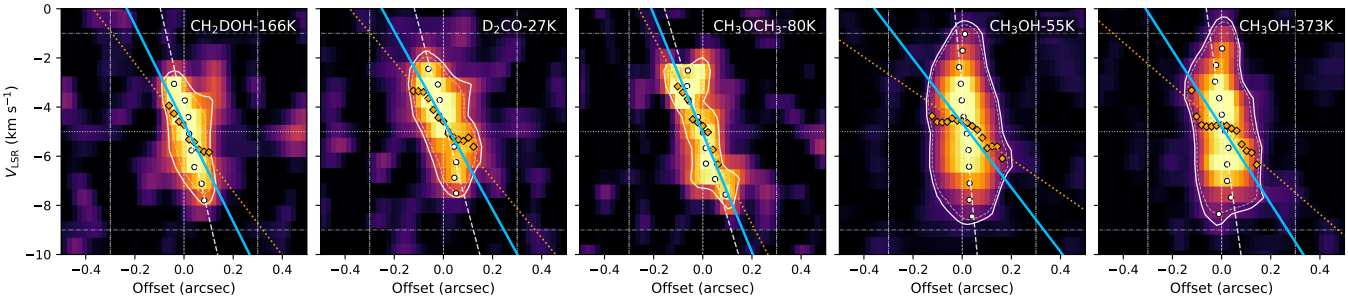


Figure B.1. Ridge-slope diagnostics for the five molecular-line PV diagrams of Figure 5. In each panel, circles show intensity-weighted offset centroids per velocity channel and squares show intensity-weighted velocity centroids per offset column, using pixels above 4$\sigma_{\rm PV}$ within $|x| < 0''.3$ and $|V - V_{\rm sys}| < 4$ km s^{-1}. The two ordinary least-squares fits (steep- and shallow-biased) and their adopted bisector (Isobe et al. 1990) are overplotted; the resulting slopes and dynamical masses are listed in Table B.1.

Table B.1. PV ridge slopes and dynamical masses.

Transition	$N_{\rm row}$	$\sigma_{\rm PV}$ (Jy beam^{-1})	α_v (km s^{-1} arcsec^{-1})	α_x (km s^{-1} arcsec^{-1})	$\alpha_{\rm bis}$ (km s^{-1} arcsec^{-1})	$M_{\rm enc}$ ($M_\odot$)
CH$_2$DOH-166K	8	1.46×10^{-3}	-42.96 ± 24.12	-13.04 ± 5.15	-20.02 ± 19.39	0.32 ± 0.91
D$_2$CO-27K	9	1.45×10^{-3}	-37.62 ± 16.61	-11.93 ± 4.11	-18.13 ± 15.43	0.26 ± 0.63
CH$_3$OCH$_3$-80K	9	1.26×10^{-3}	-33.22 ± 14.30	-19.03 ± 8.36	-24.20 ± 10.91	0.46 ± 0.51
CH$_3$OH-55K	12	1.57×10^{-3}	-108.06 ± 37.98	-6.93 ± 2.96	-13.08 ± 54.03	0.13 ± 3.42
CH$_3$OH-373K	11	1.44×10^{-3}	-70.93 ± 46.34	-8.74 ± 4.96	-15.60 ± 38.86	0.19 ± 2.15

$N_{\rm row}$ is the number of velocity channels used in the fit and $\sigma_{\rm PV}$ the rms noise of each PV diagram. α_v and α_x are the least-squares slopes fitted to the per-channel offset centroids and the per-column velocity centroids, and $\alpha_{\rm bis}$ is their bisector (Isobe et al. 1990), adopted as the final slope. $M_{\rm enc}$ is the enclosed mass implied by $\alpha_{\rm bis}$ at $R = 0''.1$ (Section 3.3).

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